

Cross-dataset 1 Results and Analysis

We train and test our dataset for:

GRU (Gated Recurrent Unit), Transformer and GAN (Generative Adversarial Network)

Train Dataset: orig_mod_timecorr.csv

Test Dataset: samedata_r2_updated.txt

Summary:

For this dataset, model performance varies significantly between voltage and power tasks. For voltage, all models perform poorly with negative R^2 though Transformer is relatively better than GRU, while GAN fails completely. For power, Transformer performs best (R^2 approx. 0.81), followed by GRU with moderate performance, whereas GAN again performs very poorly. This indicates that attention-based models generalize better on this dataset, while temporal models (GRU) struggle and GAN remains unsuitable. **Overall, Transformer is the most reliable model for this dataset.**

Voltage GRU Results:

MAE: 0.5213884068014274

RMSE: 0.5822451475970581

R2: -2.379852387960277

Power GRU Results:

MAE: 26.434416990774185

RMSE: 112.4893431264364

R2: 0.34890340999560054

Voltage Transformer Results:

MAE: 0.2917532645549319

RMSE: 0.40883431868899495

R2: -0.6660976557050153

Power Transformer Results:

MAE: 31.936079560661955

RMSE: 60.27891880702255

R2: 0.8130334391616596

Voltage GAN Results:

MAE: 1.2403887431572886

RMSE: 1.5336175069165185

R2: -22.4444408537559

Power GAN Results:

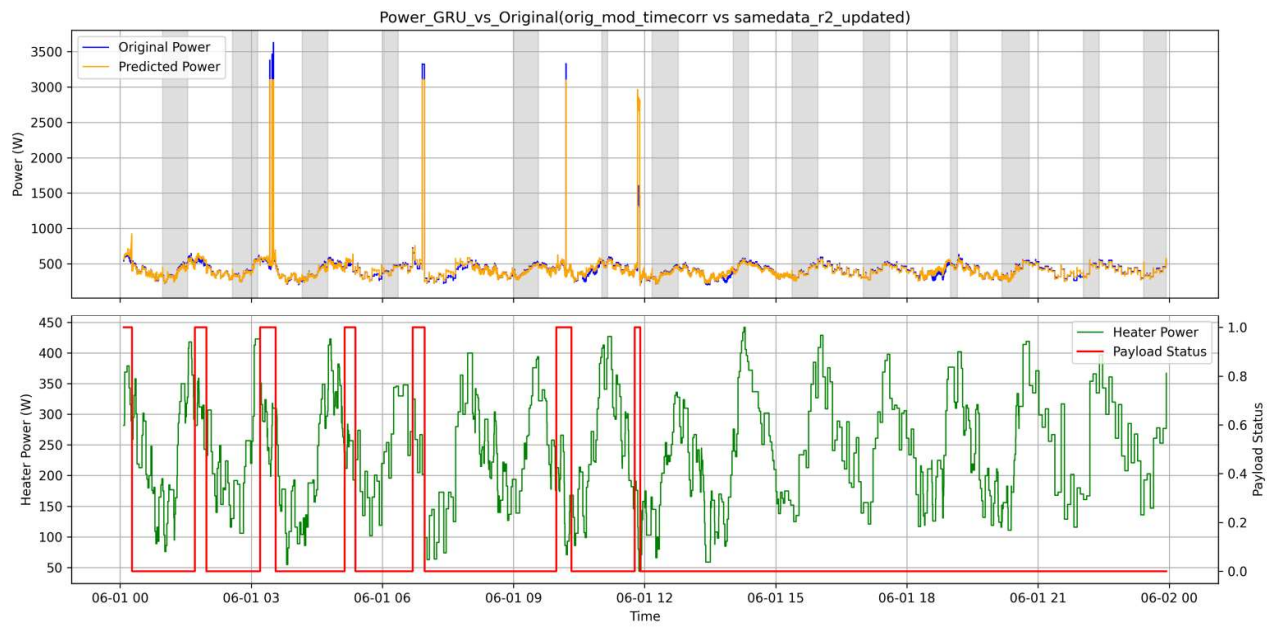
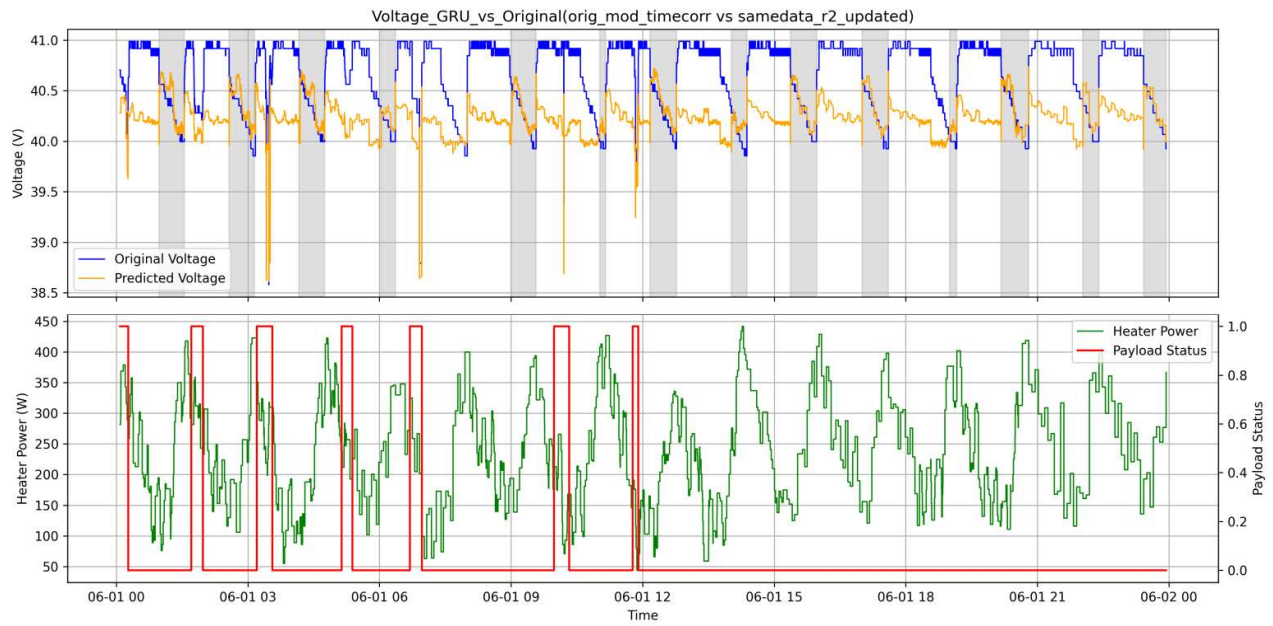
MAE: 188.53909377814273

RMSE: 231.39496476330186

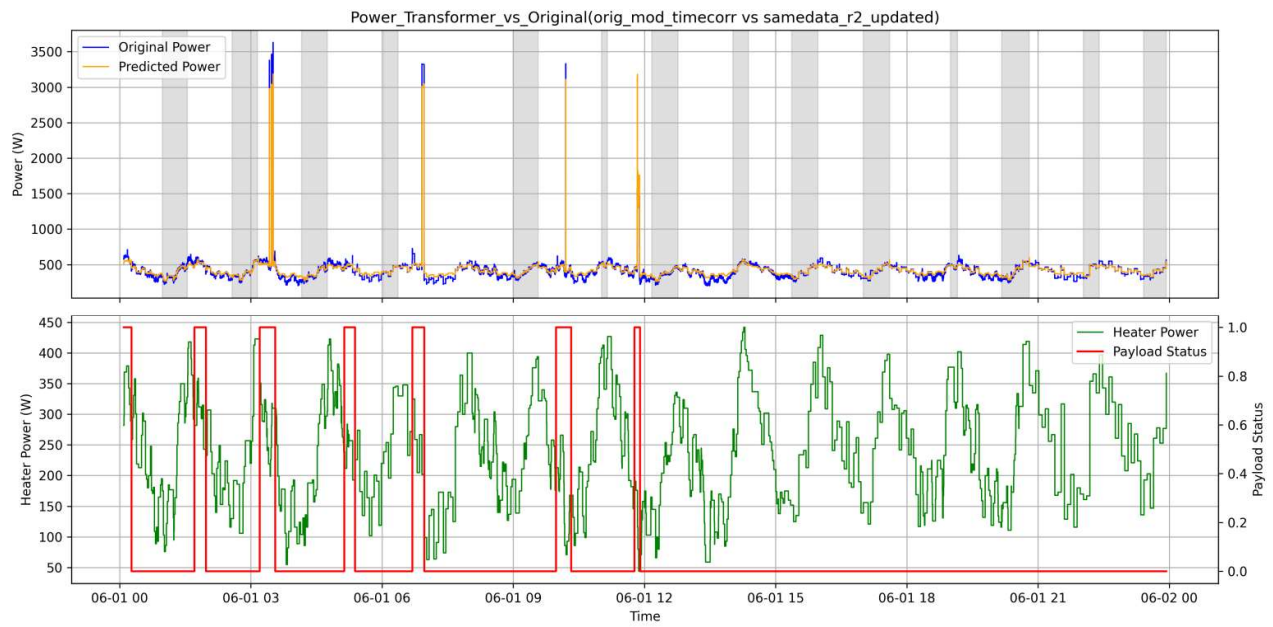
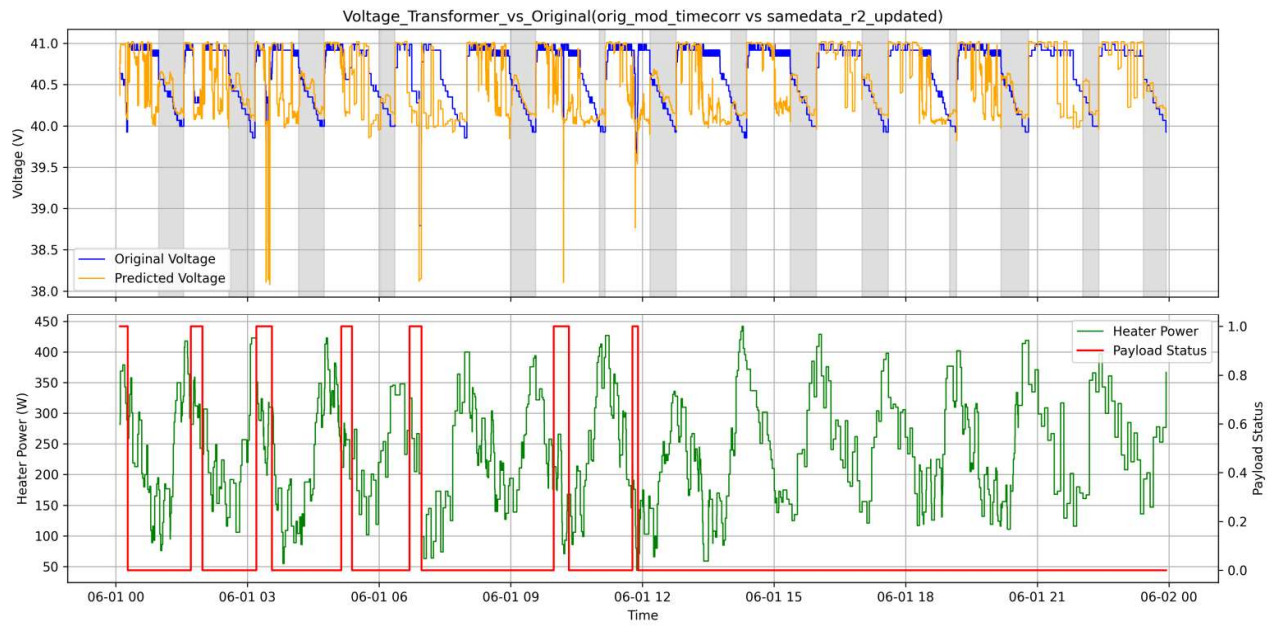
R2: -1.7551220345448488

Plots:

1. GRU



2. Transformer



3. GAN

